



Bhavesh Gurnani

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ACADEMIC DETAILS:

YEAR	DEGREE / EXAM	INSTITUTE	CGPA / MARKS (%)
2026*	Dual (B. Tech + M. Tech)	Indian Institute of Technology, Delhi	8.50(Out of 10)
2021	Class XII	Lord Buddha Public School, Kota	84.00%
2019	Class X	Lord Buddha Public School, Kota	94.00%

SCHOLASTIC ACHIEVEMENTS

- Joint Entrance Examination (JEE) Advanced:** Secured All India Rank of **116** among 200 thousand students. [2021]
- KVPY Fellowship:** Accorded to the prestigious scholarship by IISc and secured an All India Rank 302. [2021]
- NTSE Scholar:** Accorded to the Prestigious scholarship by NCERT and was among top 2000 students. [2019]
- INMO Qualifier:** Among top 300 students of country to qualify for Indian National Mathematics Olympiad. [2019]
- Joint Entrance Examination (JEE) Mains:** Secured All India Rank **493** among **1 million +** students. [2021]
- Achieved Top 50** selection twice in the **Tower Research Limestone Data Challenge**. [2024]

INTERNSHIPS

- Quantitative Researcher Intern | Ebullient Securities | Gurugram** [May 2025 – July 2025]
 - Built and back tested alpha signals for short-term equity directionality from equities and futures.
 - Automated alpha-generation pipeline using LLMs with custom evaluation loops for iterative refinement
- Software/Research Engineer Intern | Compiler AI Labs Private Limited | New Delhi** [May 2024 - July 2024]
 - Implemented MISRA C static analysis rules within Clang compiler across Preprocessor, AST, and LLVM stages.
 - Enhanced Equivalence Checker tool for formal verification of compiled executable code against C source code.

TECHNICAL SKILLS

- Programming Languages** : Proficient in C, C++, Java, python, VHDL, SML, Prolog, ml-lex, ml-yacc
- Artificial Intelligence and ML** : Tensorflow, Scikit-learn, NumPy, pandas, open cv, Py torch

ENGINEERING COURSES

- Computer Science Courses:** Digital Logic and Systems, Data Structures and Algorithms, Discrete Mathematics, Introduction to Computer Science, Computer Architecture, Programming Languages, Machine Learning, Principles of AI, Computer Networks, Analysis and design of Algorithms, Parallel and distributed Programming, Automata and Theory of Computation, Operating Systems, Database Management Systems, Natural Language Processing, Deep Learning, Logic for Computer Science, Algorithmic Graph Theory, Numerical Algorithms, Data Mining, Computer Vision
- Mathematics Courses:** Probability theory and Stochastic Process, Calculus, Linear Algebra and Differential Equations, Group Theory, Introduction to Game Theory, Algebra

ACADEMIC PROJECTS

- Improving Dialogue Systems for Medical Diagnosis** (B. Tech Project under Prof. Mausam, July 2024 – May 2025)
 - Modeled large-scale medical text data as Bayesian Networks to capture causal disease–symptom relationships.
 - Designed novel inference techniques for Bayesian Networks, improving performance over standard RAG systems.
- Expert Iteration for the Alternating Markov Games** (Independent Project, Aug 2023 - Oct 2023)
 - Implemented Expert Iteration Algorithm using Monte Carlo Tree Search (MCTS) Algorithm in the Hex game.
 - Also, Tested It on games like rollerball and other chess variations and compared its performance to Iterative deepening mini-max algorithm by making custom features and trained weights of features using imitation learning.
- Lexer, Parser, Interpreter of Language supporting rational numbers** (COL226, Prof. S. Arun Kumar, April 2023)
 - Built custom BigInt and Rational module in SML and then used ML-Lex and ML-Yacc to generate Abstract Syntax Tree (AST) and then built various maps to implement static scoping and finally an evaluator that does DFS on AST.
 - Support for variable declaration, conditionals, loops, **Recursions**, BigInt, rational & logical operations and input-output operations.
- Custom Multi-Threading Library** (COP290, Prof. Abhilash Jindal, Feb 2023 - March 2023)
 - Used ucontext library in C to build our own custom multithreading library and used it to test and simulate parallel threads
 - Tested it on calculating word count of each word present in different files of given directory using HashMap and compared its performance to built-in pthread library by varying various factors like words frequency, files number etc.

Extra-Curricular Activities

- Teaching Assistant for AI course under Prof. Mausam:** Designed quizzes and Assignments for COL333 [2025]
- Took ML in Astronomy Workshop; built LSTM-Attention models on light curves for object classification [Code]. [2024]
- ARIES Research Team Member:** Conducted NLP sessions and contributed to multiple industry research projects. [2024]
- Led **Inter IIT Tech** Team for Adobe Research PS; designed solution for artifacts detection in AI-generated images [2024]
- Academic Mentor:** Mentored 150+ freshers for MTL100(Calculus) Course by taking doubt sessions in a hybrid mode [2023]